

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 641 Computer Graphics — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
GAMES101 — Lingqi Yan, Introduction to Modern Computer Graphics (free, English slides)	Primary lecture series. Clear, modern, and closely aligned to this course's order. Watch the mapped lecture before each module.
CMU 15-462/662 Computer Graphics — Keenan Crane (free video)	Second voice, more mathematical. Best for geometry, sampling, and the parts where you want the theory tightened.
Cem Yuksel — Interactive Computer Graphics, Utah (free video)	Best practical treatment of the GPU pipeline and the API layer. Use from Module 07 onward.
UC San Diego CSE167x Computer Graphics (edX, free audit)	Structured assignments with automated feedback. Good if you want graded checkpoints.
Scratchapixel — graphics from scratch (free)	Written reference with full derivations. The place to go when a lecture moved too fast.
LearnOpenGL (free)	The API tutorial. Use for mechanics only — the theory comes from this course, not from here.

Module-by-module map

Module 01 — The Graphics Pipeline, End to End

Resource	What it covers for this module
GAMES101 Lecture 01 — Overview of Computer Graphics	The same framing of what graphics is and why. Watch before this module.
CMU 15-462 Lecture 01 — Course Introduction	Crane's framing of graphics as a sampling-and-representation problem is unusually clear and complements this module directly.
Cem Yuksel — Introduction to Computer Graphics, Lecture 1	The most hardware-grounded overview of the three. Good second pass.
Scratchapixel — 'An Overview of the Rasterization Algorithm'	Written, with code, covering the twelve-line pipeline in detail.
LearnOpenGL — Hello Triangle	Mechanics only: how the stages appear in an actual API. Do not take the theory from here.

Module 02 — Vectors, Frames, and Homogeneous Coordinates

Resource	What it covers for this module
3Blue1Brown — Essence of Linear Algebra	Chapters 1–9. The best available geometric intuition for bases, determinants, and change of basis. Watch before this module if linear algebra feels like symbol-pushing.
GAMES101 Lecture 02 — Review of Linear Algebra	The same material aimed specifically at graphics, including the geometric meaning of both products.
Scratchapixel — Geometry	Written reference with the normal-matrix derivation in full.
Immersive Math — interactive linear algebra	Manipulable figures. Useful for building intuition about frames and projections.

Module 03 — Transformations: Model, View, Projection

Resource	What it covers for this module
GAMES101 Lectures 03–04 — Transformation, Viewing	The clearest derivation of the perspective matrix available free, including the similar-triangles argument and the squish-then-ortho view.
Scratchapixel — The Perspective and Orthographic Projection Matrix	Full written derivation with every entry justified. The reference to reach for when a step in the derivation does not land.
LearnOpenGL — Coordinate Systems	The API mechanics: where each matrix goes and how to upload it.
NVIDIA — Depth Precision Visualized	The definitive treatment of depth precision and reversed-Z, with diagrams that make the $1/z$ distribution obvious.

Module 04 — Rasterization: From Triangles to Fragments

Resource	What it covers for this module
GAMES101 Lectures 05–06 — Rasterization	Coverage, the inside test, and the introduction to sampling that sets up Module 09.
Fabian Giesen — 'A trip through the Graphics Pipeline' and the rasterization series	The best free writing on how real hardware rasterizes: fill rules, fixed point, hierarchical traversal. Essential reading for this module.
Scratchapixel — Rasterization: a Practical Implementation	Full working code including the perspective-correction derivation.
CMU 15-462 — Drawing a Triangle / Sampling	Frames coverage as a sampling problem, which is the framing Module 09 builds on.

Module 05 — Visibility: Depth, Culling, and the Z-Buffer

Resource	What it covers for this module
GAMES101 Lecture 07 — Shading 1 (Z-buffering section)	The z-buffer and painter's algorithm comparison, including the cyclic overlap case.
NVIDIA — Depth Precision Visualized	The authoritative treatment of z-fighting and reversed-Z, with the diagrams that make the $1/z$ distribution obvious.
Fabian Giesen — 'A trip through the Graphics Pipeline', early-Z and hierarchical Z parts	How Hi-Z, early-Z, and depth compression actually work in hardware, and exactly what turns them off.
McGuire & Bavoil — Weighted Blended Order-Independent Transparency	Open-access paper. The practical OIT method, with code.

Module 06 — Light, Colour, and Local Shading

Resource	What it covers for this module
GAMES101 Lectures 07–09 — Shading	Lambert, Blinn–Phong, and the three shading frequencies, with the cosine-law derivation done geometrically.
John Hable — 'Why a Filmic Curve' and the gamma articles	The clearest practical writing on linear workflow and tonemapping, from someone who shipped it.
LearnOpenGL — Basic Lighting and Gamma Correction	The API mechanics of sRGB textures and framebuffers, with before/after images.
Naty Hoffman — 'Physics and Math of Shading' (SIGGRAPH course, free)	The best free introduction to radiometry for graphics. Read it before Module 11.

Module 07 — The Programmable Pipeline: Shaders in Practice

Resource	What it covers for this module
Cem Yuksel — Interactive Computer Graphics (Utah), GPU pipeline lectures	The best free treatment of the programmable pipeline and the API layer. Primary source from this module onward.
LearnOpenGL — Getting Started and Lighting sections	Mechanics: context creation, buffers, shader compilation, uniforms. Use for the API, not for the theory.
Fabian Giesen — 'A trip through the Graphics Pipeline'	What the hardware is actually doing at each stage. The best free writing on the subject.
RenderDoc documentation and quick-start	Work through this once with a real capture. It pays for itself within the week.
The Book of Shaders	Fragment-shader intuition through small interactive examples. Good for building fluency in the language itself.

Module 08 — Texture Mapping, Filtering, and Mipmaps

Resource	What it covers for this module
GAMES101 Lectures 09–10 — Texture Mapping, Advanced Texturing	Mipmapping derived as a sampling fix, with excellent diagrams of the anisotropic footprint problem.
LearnOpenGL — Textures, Normal Mapping, Parallax Mapping	Complete working TBN construction including the handedness detail.
Scratchapixel — Texture Mapping	Written treatment with the mip-level derivation in full.
Nathan Reed — 'Understanding BCn Texture Compression Formats'	Why normal maps use BC5 and colour uses BC7, and what each format sacrifices. Short, practical, free.

Module 09 — Sampling, Aliasing, and Antialiasing

Resource	What it covers for this module
GAMES101 Lectures 06 and 09 — Antialiasing, Texture aliasing	Aliasing presented properly in the frequency domain, with the spatial/frequency duality made visual.
CMU 15-462 — Sampling and Aliasing	The most rigorous free treatment: Fourier analysis applied directly to the rendering case.
Jorge Jimenez et al. — SMAA and 'Filmic SMAA' (SIGGRAPH course, free)	Post-process antialiasing done well, with an honest account of its limits.
Brian Karis — 'High Quality Temporal Supersampling' (SIGGRAPH course, free)	The reference presentation on practical TAA, including neighbourhood clamping. Still the clearest explanation of why TAA ghosts.

Module 10 — Shadows

Resource	What it covers for this module
GAMES101 Lecture 10 — Shadow Mapping	The algorithm and the bias problem, clearly derived.
LearnOpenGL — Shadow Mapping and Point Shadows	Working implementation including the cube-map point-light case.
Microsoft — 'Common Techniques to Improve Shadow Depth Maps'	The definitive free reference on cascades, bias strategies, and texel snapping. Read it in full before implementing cascades.
Fernando — Percentage-Closer Soft Shadows (NVIDIA, free)	The original PCSS formulation with the blocker-search derivation.

Module 11 — Physically Based Shading and Image-Based Lighting

Resource	What it covers for this module
Brian Karis — 'Real Shading in Unreal Engine 4' (SIGGRAPH course, free)	The split-sum approximation in the author's own words, with the practical shortcuts. The single most important reference for this module.
Sébastien Lagarde & Charles de Rousiers — 'Moving Frostbite to PBR' (free)	The most complete free account of a production PBR pipeline, including all the details other papers omit.
LearnOpenGL — PBR Theory, Lighting, and IBL	Complete working implementation of everything in this module, including generating the irradiance map, prefiltered map, and BRDF LUT.
Naty Hoffman — 'Physics and Math of Shading' (free)	The derivation of microfacet theory from first principles. Read this if the D, F, G decomposition feels arbitrary.
Physically Based Rendering, 4th ed. — free online	Chapter 9 on reflection models. The rigorous treatment, and the bridge to CSCE 647.

Module 12 — Geometry: Meshes, Curves, and Tessellation

Resource	What it covers for this module
Keenan Crane — Discrete Differential Geometry (free notes, video, code)	The proper treatment of mesh processing and geometry. This is the primary text for CSCE 645; start it here.
GAMES101 Lectures 11–12 — Geometry	Curves, surfaces, subdivision, and simplification at the right level for this module.
libigl tutorial (free)	Working code for half-edge traversal, simplification, smoothing, and parameterisation. Read the source.
Garland & Heckbert — 'Surface Simplification Using Quadric Error Metrics' (free)	The original paper. Short, clear, and still the basis of every production simplifier.
Brian Karis — 'Nanite: A Deep Dive' (SIGGRAPH, free)	How virtualised geometry actually works. Read after the rest of this module.

Module 13 — Animation: Hierarchies, Skinning, and Interpolation

Resource	What it covers for this module
GAMES101 Lectures 19–20 — Animation	Keyframing, rigging, and skinning at the right level, with the rotation-interpolation problem shown clearly.
3Blue1Brown & Ben Eater — 'Visualizing quaternions' (interactive)	The best available intuition for why quaternions represent rotations. Interactive and free.
Kavan et al. — 'Skinning with Dual Quaternions' (free)	The paper that fixed candy-wrapping, with the derivation and the honest account of the bulging trade-off.
glTF 2.0 specification — skins and animations	The standard runtime format. Reading the skinning section is the fastest way to see exactly what data a real pipeline ships.
Jason Gregory — Game Engine Architecture, animation chapter (sample chapters free)	How animation systems are actually structured in production engines.